

Ali Aqdas

Ph.D. Computer Science, Purdue University
Security, Correctness and Reliability of Systems

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RESEARCH PROFILE

Computer Systems Architect focused on the security, microarchitectural reliability, and correctness of next-generation infrastructures, leveraging a versatile hardware-software stack to solve complex systems challenges.

EDUCATION

Purdue University Ph.D. in Computer Science Advisor: Dr. Kazem Taram Thesis: Microarchitectural Security in Compute Express Link (CXL) Systems	Aug 2024 – Present West Lafayette, IN
National University of Sciences and Technology (NUST) Bachelor of Electrical Engineering CGPA: 3.93/4.00 Capstone: SoC Implementation of RISC-V Vector Processor (RVV 1.0)	Sept 2019 – Jun 2023 Islamabad, Pakistan

TECHNICAL EXPERTISE

- **Languages & ISAs:** C, C++, Python, Bash; x86-64, ARM64, RISC-V Assembly
- **Hardware & Systems Architecture:** Compute Express Link (CXL 2.0/3.0), Cache Coherence (MESI/MESIF), Disaggregated Memory, RDMA
- **RTL Design & ASIC Flow:** FPGA, Verilog/SystemVerilog, TSMC 22nm/65nm, ASAP7
- **Hardware Security & Crypto:** Microarchitectural Timing Attacks, Hardware Private Circuits (HPC), Side-Channel Analysis/Mitigation, TEEs
- **Systems Software & Tools:** Linux Kernel/Environment, Ghidra, Git, GDB, Make

PUBLICATIONS

Compute eXpress Leak: New Cache Attacks Using CXL Devices	MICRO 2026
Ali Aqdas, Zheng Zhang, Yue Pan, Zixuan Wang, Tajana Rosing, Jishen Zhao and Kazem Taram	

RESEARCH EXPERIENCE

SecArch Lab, Purdue University Graduate Research Assistant	Mar 2025 – Present West Lafayette, IN
■ CXL Security: Discovered protocol-level cache coherence vulnerabilities in CXL Type-2/Type-3 shared memory topologies to design device-initiated (PUSH+PUSH) and cross-host (SnoopDog) side-channel attacks; built a full-stack verification platform (Linux driver + FPGA RTL) achieving ~17 kbps covert channels on physical hardware alongside a QEMU/KVM simulation framework for SnoopDog, and engineered a hardware-validated constant-time mitigation protocol with < 3.13% performance overhead. ■ Hardware Private Circuits: Hardened a pipelined AES core with masked primitives from SMAesH; executed RTL design flows and evaluated PPA overheads.	
Next-G Architectures Lab, Purdue University Graduate Research Assistant	Aug 2024 – Mar 2025 West Lafayette, IN
■ RoCE-based Transport for Distributed ML: Developed a simulation testbed by integrating RDMA (CoyoteOS) with SDNet framework for characterizing and optimizing transport layers to eliminate protocol overhead and mitigate network bottlenecks in distributed training clusters.	
SoC Design Lab, NUST Research Assistant	Jun 2022 – Jul 2024 Islamabad, Pakistan
■ Led the RTL design and integration of a RISC-V Vector Processor (RVV 1.0) with a custom scalar core. ■ Executed the complete ASIC synthesis and floorplanning flow (TSMC 65nm/22nm, ASAP7) using Cadence Genus and Innovus.	

- Analyzed **PPA tradeoffs** (Power, Area, and Timing) and optimized microarchitectural components for high-bandwidth vector operations.
- Developed vectorized kernels using intrinsic instructions for performance benchmarking and architectural verification.

TEACHING EXPERIENCE

Purdue University

Graduate Teaching Assistant

Aug 2024 – Present

West Lafayette, IN

- **CS-250: Computer Architecture** (Fall 2024, Spring 2025, Fall 2025).
- Responsible for leading lab sessions, developing course materials, and assisting students with low-level systems programming and architectural concepts.

NUST Chip Design Center

Instructor

May 2024 – July 2024

Islamabad, Pakistan

- Designed and conducted hands-on technical labs for training industry recruits in ASIC design flows.
- Collaborated with faculty to develop advanced training modules for RISC-V microarchitecture.

SERVICE & HONORS

Artifact Evaluator, ISCA 2025–2026: Evaluated reproducibility and technical rigor of research artifacts for the International Symposium on Computer Architecture.

NUST Scholarship Award: Received for top academic performance across all four years of undergraduate study (2019–2023).

Academic Scholarship: Awarded for excellence in Higher Secondary and Secondary Education (2015–2019).

SELECTED PROJECTS

XOR-Memory Generator: Developed a software tool in C++ to generate parameterizable Verilog files for multi-ported memory systems on FPGAs, enabling high-bandwidth parallel access.

TinyML Fall Detection System: Engineered and deployed an optimized CNN using TensorFlow Lite Micro on ARM Cortex-M microcontrollers for real-time edge inference.

Bresenham Circle Algorithm on FPGA: Implemented circle-drawing logic in Verilog with VGA output on a DE1-SoC platform.

Angle of Arrival Estimation: Simulated the MUSIC algorithm in MATLAB and implemented it on FPGA for real-time signal processing in the 4–6 GHz range.